Class 12 Mathematics

Chapter 2: Inverse Trigonometric Functions

Practice Set A: step-wise solutions

20 questions **43** marks **3** levels

Each row is one line of working. The red number in the right margin is what an examiner gives for that line, as in the CBSE marking scheme (½ is half a mark). Write every line that carries a mark. ✓ ½

Level 1

8 marks

1 **MCQ** PYQ-style — 1 mark

The principal value of $\cot^{-1}\left(-\frac{1}{\sqrt{3}}\right)$ is:

(A) $-\frac{\pi}{3}$ **(B)** $\frac{\pi}{3}$ **(C)** $\frac{2\pi}{3}$ **(D)** $\frac{5\pi}{6}$

Formula / idea: Principal value branch of $\cot^{-1} x$ is $(0, \pi)$.

1 Write the range of the principal value branch.
$\cot^{-1} x \in (0, \pi)$

2 Recall the standard value.
$\cot \frac{\pi}{3} = \frac{1}{\sqrt{3}}$

3 Find the angle in $(0, \pi)$ whose cot is $-\frac{1}{\sqrt{3}}$. ✓ 1
$\cot\left(\pi - \frac{\pi}{3}\right) = -\cot\frac{\pi}{3} = -\frac{1}{\sqrt{3}}$

4 Since $\frac{2\pi}{3} \in (0, \pi)$, the principal value is $\frac{2\pi}{3}$. Option (C).

Answer **(C)** $\frac{2\pi}{3}$

Watch out: Writing $-\frac{\pi}{3}$: it is not in $(0, \pi)$, the range of $\cot^{-1} x$.

2 MCQ PYQ-style 1 mark

The value of $\cos^{-1}\left(\sin\frac{4\pi}{3}\right)$ is:

(A) $\frac{4\pi}{3}$ **(B)** $-\frac{\pi}{6}$ **(C)** $\frac{5\pi}{6}$ **(D)** $\frac{\pi}{6}$

Formula / idea: $\cos^{-1} y$ is the angle in $[0,\pi]$ whose cosine is y.

1 Find $\sin\frac{4\pi}{3}$.

$\sin\frac{4\pi}{3} = \sin\left(\pi+\frac{\pi}{3}\right) = -\sin\frac{\pi}{3} = -\frac{\sqrt{3}}{2}$

2 Find the angle in $[0,\pi]$ whose cosine is $-\frac{\sqrt{3}}{2}$. ✓ 1

$\cos\frac{5\pi}{6} = -\frac{\sqrt{3}}{2}, \quad \frac{5\pi}{6} \in [0,\pi]$

3 So $\cos^{-1}\left(\sin\frac{4\pi}{3}\right) = \frac{5\pi}{6}$. Option (C).

Answer **(C)** $\frac{5\pi}{6}$

Watch out: Writing $\frac{4\pi}{3}$: $\cos^{-1}$ and $\sin$ do not cancel, and $\frac{4\pi}{3}$ is outside $[0,\pi]$.

3 MCQ CBSE SQP 2025-26 1 mark

The inverse trigonometric function whose domain is $\left[-\frac{1}{3},\frac{1}{3}\right]$ is:

(A) $\cos^{-1} x$ **(B)** $\cos^{-1}\left(\frac{x}{3}\right)$ **(C)** $\cos^{-1}(3x)$ **(D)** $3\cos^{-1} x$

Formula / idea: $\cos^{-1} t$ is defined only for $-1 \le t \le 1$.

1 Domain of $\cos^{-1} x$ and of $3\cos^{-1} x$.

$-1 \le x \le 1 \Rightarrow [-1,1]$

2 Domain of $\cos^{-1}\left(\frac{x}{3}\right)$.

$-1 \le \frac{x}{3} \le 1 \Rightarrow [-3,3]$

3 Domain of $\cos^{-1}(3x)$. ✓ 1

$-1 \le 3x \le 1 \Rightarrow -\frac{1}{3} \le x \le \frac{1}{3}$

4 Only $\cos^{-1}(3x)$ has domain $\left[-\frac{1}{3}, \frac{1}{3}\right]$. Option (C).

Answer **(C)** $\cos^{-1}(3x)$

Watch out: Choosing $\cos^{-1}\left(\frac{x}{3}\right)$: dividing x by 3 stretches the domain to $[-3, 3]$.

4 **MCQ** CBSE SQP 2024-25 1 mark

The graph drawn below depicts:



(A) $y = \sin^{-1} x$

(B) $y = \cos^{-1} x$

(C) $y = \operatorname{cosec}^{-1} x$

(D) $y = \cot^{-1} x$

Formula / idea: Read the domain and the range from the graph, then match the principal value branch.

1 Read the domain from the graph.
$x \in [-1, 1]$

2 Rule out $\operatorname{cosec}^{-1} x$ (domain $\mathbb{R} - (-1, 1)$) and $\cot^{-1} x$ (domain $\mathbb{R}$).

3 Read the range: from π at $x = -1$ down to 0 at $x = 1$.
$y \in [0, \pi]$

4 Range $[0, \pi]$ belongs to $\cos^{-1} x$; $\sin^{-1} x$ has range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$. ✓ 1

5 Hence the graph is $y = \cos^{-1} x$. Option (B).

Answer **(B)** $y = \cos^{-1} x$

Watch out: Picking $\sin^{-1} x$ because the domain is $[-1, 1]$; the range $[0, \pi]$ decides it.

5 MCQ PYQ-style 1 mark

The principal value of $\cos^{-1}\left(-\frac{1}{2}\right)+2\sin^{-1}\left(\frac{1}{2}\right)$ is:

(A) 0 **(B)** $\frac{5\pi}{6}$ **(C)** π **(D)** $\frac{4\pi}{3}$

Formula / idea: Ranges: $\cos^{-1}x\in[0,\pi]$, $\sin^{-1}x\in\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$.

1 Angle in $[0,\pi]$ with cosine $-\frac{1}{2}$.
$\cos\frac{2\pi}{3}=-\frac{1}{2}\Rightarrow\cos^{-1}\left(-\frac{1}{2}\right)=\frac{2\pi}{3}$

2 Angle in $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$ with sine $\frac{1}{2}$.
$\sin^{-1}\left(\frac{1}{2}\right)=\frac{\pi}{6}$

3 Add.
$\frac{2\pi}{3}+2\cdot\frac{\pi}{6}=\frac{2\pi}{3}+\frac{\pi}{3}=\pi$ ✓ 1

4 Hence the value is π. Option (C).

Answer **(C)** π

Watch out: Taking $\cos^{-1}\left(-\frac{1}{2}\right)=-\frac{\pi}{3}$, which gives 0; negative angles are outside $[0,\pi]$.

6 Assertion-Reason CBSE SQP 2025-26 1 mark

Assertion (A): The value of the expression $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)+\tan^{-1}1-\sec^{-1}\left(\sqrt{2}\right)$ is $\frac{\pi}{4}$. **Reason (R):** The principal value branch of $\sin^{-1}x$ is $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$ and that of $\sec^{-1}x$ is $[0,\pi]-\left\{\frac{\pi}{2}\right\}$.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) **Assertion (A) is false, but Reason (R) is true.**

Formula / idea: Each principal value must lie in the range of its principal value branch.

1 Find the three principal values.
$\sin^{-1}\frac{\sqrt{3}}{2}=\frac{\pi}{3},\quad\tan^{-1}1=\frac{\pi}{4},\quad\sec^{-1}\sqrt{2}=\frac{\pi}{4}$

2	Check A: the value is $\frac{\pi}{3}$, not $\frac{\pi}{4}$, so A is false. $\frac{\pi}{3}+\frac{\pi}{4}-\frac{\pi}{4}=\frac{\pi}{3}$	
3	Check R: both branches are stated correctly, so R is true.	
4	A is false, so R cannot explain A.	
5	A is false and R is true. Option (D).	✓ 1

Answer **(D) Assertion (A) is false, but Reason (R) is true.**

Watch out: Cancelling $\tan^{-1}1$ with $\sec^{-1}\sqrt{2}$ and then taking $\sin^{-1}\frac{\sqrt{3}}{2}=\frac{\pi}{4}$.

7 **Short answer** CBSE 2023 2 marks

Evaluate: $\sin^{-1}\left(\sin\frac{3\pi}{4}\right)+\cos^{-1}(\cos\pi)+\tan^{-1}(1)$.

Formula / idea: $\sin^{-1}(\sin\theta)=\theta$ only when $\theta\in\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$; $\cos^{-1}(\cos\theta)=\theta$ only when $\theta\in[0,\pi]$.

1	$\frac{3\pi}{4}$ is not in $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$, so rewrite the sine. $\sin\frac{3\pi}{4}=\sin\left(\pi-\frac{3\pi}{4}\right)=\sin\frac{\pi}{4}$	
2	Since $\frac{\pi}{4}\in\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$: $\sin^{-1}\left(\sin\frac{3\pi}{4}\right)=\sin^{-1}\left(\sin\frac{\pi}{4}\right)=\frac{\pi}{4}$	✓ 1
3	$\pi\in[0,\pi]$, and $\tan\frac{\pi}{4}=1$. $\cos^{-1}(\cos\pi)=\pi,\qquad\tan^{-1}(1)=\frac{\pi}{4}$	✓ ½
4	Add the three values. $\frac{\pi}{4}+\pi+\frac{\pi}{4}=\frac{3\pi}{2}$	✓ ½

Answer $\frac{3\pi}{2}$

Watch out: Writing $\sin^{-1}\left(\sin\frac{3\pi}{4}\right)=\frac{3\pi}{4}$: that angle is outside the range of $\sin^{-1}$.

Level 2

16 marks

8 **Short answer** CBSE 2023 2 marks

Evaluate: $3\sin^{-1}\left(\frac{1}{\sqrt{2}}\right)+2\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)+\cos^{-1}(0)$.

Formula / idea: Use the principal value of each term, then add.

1	$\sin\frac{\pi}{4}=\frac{1}{\sqrt{2}}$ and $\frac{\pi}{4}\in\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$. $3\sin^{-1}\left(\frac{1}{\sqrt{2}}\right)=3\cdot\frac{\pi}{4}=\frac{3\pi}{4}$	✓ ½
2	$\cos\frac{\pi}{6}=\frac{\sqrt{3}}{2}$ and $\frac{\pi}{6}\in[0,\pi]$. $2\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)=2\cdot\frac{\pi}{6}=\frac{\pi}{3}$	✓ ½
3	$\cos\frac{\pi}{2}=0$ and $\frac{\pi}{2}\in[0,\pi]$. $\cos^{-1}(0)=\frac{\pi}{2}$	✓ ½
4	Add, using the LCM 12. $\frac{3\pi}{4}+\frac{\pi}{3}+\frac{\pi}{2}=\frac{9\pi+4\pi+6\pi}{12}=\frac{19\pi}{12}$	✓ ½

Answer $\frac{19\pi}{12}$

Watch out: Multiplying inside the bracket, e.g. writing $\sin^{-1}\left(\frac{3}{\sqrt{2}}\right)$; the 3 multiplies the angle.

9 **Short answer** CBSE 2025 2 marks

Find the domain of $f(x)=\sin^{-1}\left(-x^2\right)$.

Formula / idea: $\sin^{-1}t$ is defined only for $-1\le t\le 1$.

1	The argument $-x^2$ must lie in $[-1,1]$. $-1\le -x^2\le 1$	✓ ½
2	Multiply by -1 and reverse the signs. $-1\le x^2\le 1$	✓ ½
3	$x^2\ge 0>-1$ always, so only $x^2\le 1$ is needed. $x^2\le 1\ \Rightarrow\ -1\le x\le 1$	✓ ½

4 Hence the domain of f is $[-1, 1]$. ✓ ½
$\text{Domain} = [-1, 1]$

Answer **Domain** $= [-1, 1]$

Watch out: Forgetting to reverse the inequality signs when multiplying by -1.

10 **Short answer** CBSE 2025 2 marks

Find the domain of $\sec^{-1}(2x + 1)$.

Formula / idea: $\sec^{-1} t$ is defined only for $t \le -1$ or $t \ge 1$, i.e. $t \in \mathbb{R} - (-1, 1)$.

1 The argument $2x + 1$ must lie outside $(-1, 1)$. ✓ ½
$2x + 1 \le -1 \quad \text{or} \quad 2x + 1 \ge 1$

2 Solve the first inequality. ✓ ½
$2x \le -2 \;\Rightarrow\; x \le -1$

3 Solve the second inequality. ✓ ½
$2x \ge 0 \;\Rightarrow\; x \ge 0$

4 Combine the two parts. ✓ ½
$\text{Domain} = (-\infty, -1] \cup [0, \infty)$

Answer **Domain** $= (-\infty, -1] \cup [0, \infty)$

Watch out: Using $-1 \le 2x + 1 \le 1$ as for $\sin^{-1}$; that is the gap where $\sec^{-1}$ is not defined.

11 **Short answer** CBSE 2024 (Comptt.) 2 marks

Evaluate: $\sin^{-1}\left(\sin\frac{13\pi}{6}\right) + \cos^{-1}\left(\cos\frac{\pi}{3}\right) + \tan^{-1}\left(\sqrt{3}\right)$.

Formula / idea: Bring each angle into the principal range before removing $\sin^{-1}(\sin\,\cdot)$ or $\cos^{-1}(\cos\,\cdot)$.

1 $\frac{13\pi}{6} = 2\pi + \frac{\pi}{6}$, and sin has period 2π. ✓ ½
$\sin\frac{13\pi}{6} = \sin\left(2\pi + \frac{\pi}{6}\right) = \sin\frac{\pi}{6}$

2 Since $\frac{\pi}{6} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$: ✓ ½
$\sin^{-1}\left(\sin\frac{13\pi}{6}\right) = \frac{\pi}{6}$

3 $\frac{\pi}{3} \in [0, \pi]$, and $\tan\frac{\pi}{3} = \sqrt{3}$. ✓ ½
$\cos^{-1}\left(\cos\frac{\pi}{3}\right) = \frac{\pi}{3}, \qquad \tan^{-1}\sqrt{3} = \frac{\pi}{3}$

4 Add the three values.
$\frac{\pi}{6}+\frac{\pi}{3}+\frac{\pi}{3}=\frac{\pi+2\pi+2\pi}{6}=\frac{5\pi}{6}$ ✓ ½

Answer $\frac{5\pi}{6}$

Watch out: Writing $\sin^{-1}\left(\sin\frac{13\pi}{6}\right)=\frac{13\pi}{6}$, which is far outside $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$.

12 **Short answer** CBSE SQP 2025-26 2 marks

Evaluate: $\tan\left(\tan^{-1}(-1)+\frac{\pi}{3}\right)$.

Formula / idea: $\tan(A-B)=\frac{\tan A-\tan B}{1+\tan A\tan B}$

1 $\tan\left(-\frac{\pi}{4}\right)=-1$ and $-\frac{\pi}{4}\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$.
$\tan^{-1}(-1)=-\frac{\pi}{4}$ ✓ ½

2 Substitute into the expression.
$\tan\left(-\frac{\pi}{4}+\frac{\pi}{3}\right)=\tan\left(\frac{\pi}{3}-\frac{\pi}{4}\right)$ ✓ ½

3 Use the formula for $\tan(A-B)$.
$=\frac{\tan\frac{\pi}{3}-\tan\frac{\pi}{4}}{1+\tan\frac{\pi}{3}\tan\frac{\pi}{4}}=\frac{\sqrt{3}-1}{\sqrt{3}+1}$ ✓ ½

4 Rationalise the denominator.
$\frac{(\sqrt{3}-1)^2}{(\sqrt{3}+1)(\sqrt{3}-1)}=\frac{4-2\sqrt{3}}{2}$
$=2-\sqrt{3}$ ✓ ½

Answer $2-\sqrt{3}$

Watch out: Taking $\tan^{-1}(-1)=\frac{3\pi}{4}$; that angle is outside $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$.

13 **Short answer** CBSE 2024 3 marks

Find the value of k if $\sin^{-1}\left[k\tan\left(2\cos^{-1}\frac{\sqrt{3}}{2}\right)\right]=\frac{\pi}{3}$.

Formula / idea: Work from the innermost bracket outwards; $\sin^{-1}t=\theta\Rightarrow t=\sin\theta$.

1 $\cos\frac{\pi}{6}=\frac{\sqrt{3}}{2}$ and $\frac{\pi}{6}\in[0,\pi]$.
$\cos^{-1}\frac{\sqrt{3}}{2}=\frac{\pi}{6}$ ✓ ½

2 Double it and take tan. ✓ 1

$\tan\left(2\cdot\frac{\pi}{6}\right)=\tan\frac{\pi}{3}=\sqrt{3}$

3 Substitute in the given equation.

$\sin^{-1}\left(\sqrt{3}\,k\right)=\frac{\pi}{3}$

4 Take sine of both sides. ✓ 1

$\sqrt{3}\,k=\sin\frac{\pi}{3}=\frac{\sqrt{3}}{2}$

5 Divide by $\sqrt{3}$. ✓ ½

$k=\frac{1}{2}$

Answer $k=\frac{1}{2}$

Watch out: Doubling inside: writing $\cos^{-1}\left(2\cdot\frac{\sqrt{3}}{2}\right)=\cos^{-1}\sqrt{3}$, which is not even defined.

14 **Short answer** CBSE 2024 3 marks

Find the value of $\sin^2\left(\cos^{-1}\frac{3}{5}\right)+\tan^2\left(\sec^{-1}3\right)$.

Formula / idea: $\sin^2\theta=1-\cos^2\theta$ and $\tan^2\theta=\sec^2\theta-1$.

1 Let $\cos^{-1}\frac{3}{5}=\alpha$. ✓ ½

$\cos\alpha=\frac{3}{5}$

2 Use $\sin^2\alpha=1-\cos^2\alpha$. ✓ 1

$\sin^2\alpha=1-\frac{9}{25}=\frac{16}{25}$

3 Let $\sec^{-1}3=\beta$. ✓ ½

$\sec\beta=3$

4 Use $\tan^2\beta=\sec^2\beta-1$. ✓ ½

$\tan^2\beta=9-1=8$

5 Add the two values. ✓ ½

$\frac{16}{25}+8=\frac{16+200}{25}=\frac{216}{25}$

Answer $\frac{216}{25}$

Watch out: Writing $\tan^2\beta=\sec^2\beta+1=10$; the identity is $\sec^2\beta=1+\tan^2\beta$.

Level 3

19 marks

15 **Long answer** CBSE 2023 3 marks

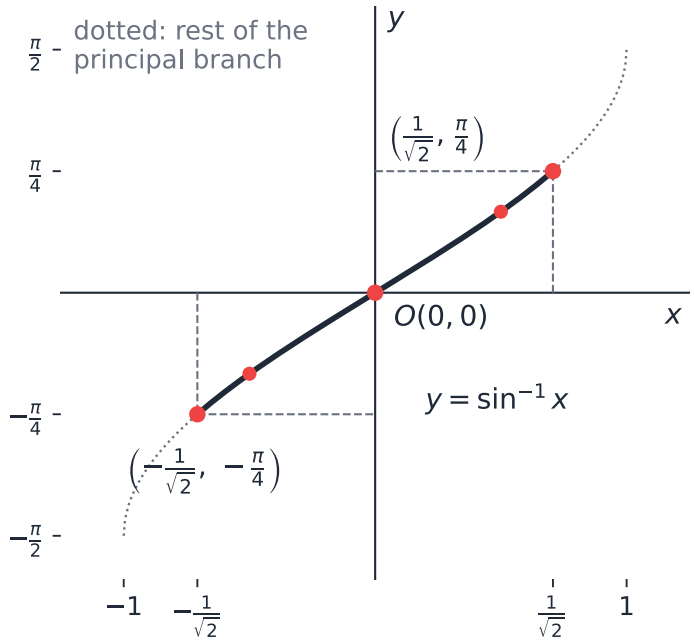
Draw the graph of $f(x) = \sin^{-1} x$, $x \in \left[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right]$. Also, write the range of $f(x)$.

Formula / idea: $y = \sin^{-1} x$ is the mirror image of $y = \sin x$, $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, in the line $y = x$; it is increasing.

1 Make a table of values from $-\frac{1}{\sqrt{2}}$ to $\frac{1}{\sqrt{2}}$. ✓ 1

x	$-\frac{1}{\sqrt{2}}$	$-\frac{1}{2}$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$
$y = \sin^{-1} x$	$-\frac{\pi}{4}$	$-\frac{\pi}{6}$	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$

2 Plot these points and join them by a smooth increasing curve. ✓ 1



3 $\sin^{-1} x$ is increasing, so the least and greatest values are at the end points.

$$f\left(-\frac{1}{\sqrt{2}}\right) = -\frac{\pi}{4}, \qquad f\left(\frac{1}{\sqrt{2}}\right) = \frac{\pi}{4}$$

4 Hence the range of f: ✓ 1

$$\text{Range} = \left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$$

Answer **Graph: the increasing arc of $y = \sin^{-1} x$ from $\left(-\frac{1}{\sqrt{2}}, -\frac{\pi}{4}\right)$ through $(0, 0)$ to $\left(\frac{1}{\sqrt{2}}, \frac{\pi}{4}\right)$; range $= \left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$.**

Watch out: Drawing the whole branch from -1 to 1 and writing range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$; the domain here is smaller.

16 **Long answer** CBSE 2024 3 marks

Find the value of $\sin^{-1}\left(-\frac{1}{2}\right)+\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)+\cot^{-1}\left(\tan\frac{4\pi}{3}\right)$.

Formula / idea: Ranges: $\sin^{-1}\in\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$, $\cos^{-1}\in[0,\pi]$, $\cot^{-1}\in(0,\pi)$.

1	$\sin\left(-\frac{\pi}{6}\right)=-\frac{1}{2}$ and $-\frac{\pi}{6}\in\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$. $\sin^{-1}\left(-\frac{1}{2}\right)=-\frac{\pi}{6}$	✓ ½
2	$\cos\frac{5\pi}{6}=-\cos\frac{\pi}{6}=-\frac{\sqrt{3}}{2}$ and $\frac{5\pi}{6}\in[0,\pi]$. $\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)=\frac{5\pi}{6}$	✓ 1
3	$\tan$ has period π. $\tan\frac{4\pi}{3}=\tan\left(\pi+\frac{\pi}{3}\right)=\tan\frac{\pi}{3}=\sqrt{3}$	✓ ½
4	$\cot\frac{\pi}{6}=\sqrt{3}$ and $\frac{\pi}{6}\in(0,\pi)$. $\cot^{-1}\left(\tan\frac{4\pi}{3}\right)=\cot^{-1}\sqrt{3}=\frac{\pi}{6}$	✓ ½
5	Add the three values. $-\frac{\pi}{6}+\frac{5\pi}{6}+\frac{\pi}{6}=\frac{5\pi}{6}$	✓ ½

Answer $\frac{5\pi}{6}$

Watch out: Writing $\cot^{-1}\left(\tan\frac{4\pi}{3}\right)=\frac{4\pi}{3}$; $\cot^{-1}$ and $\tan$ are not inverses of each other.

17 **Long answer** CBSE 2024 3 marks

Find the domain of $f(x)=\cos^{-1}\left(1-x^2\right)$. Also, find its range.

Formula / idea: $\cos^{-1}t$ is defined for $-1\le t\le 1$ and takes values in $[0,\pi]$.

1	The argument must lie in $[-1,1]$. $-1\le 1-x^2\le 1$	✓ ½
2	Subtract 1, then multiply by -1 (signs reverse). $-2\le -x^2\le 0\ \Rightarrow\ 0\le x^2\le 2$	✓ 1
3	Take square roots. $-\sqrt{2}\le x\le\sqrt{2}\ \Rightarrow\ \text{Domain}=\left[-\sqrt{2},\sqrt{2}\right]$	✓ ½

4 On this domain $1 - x^2$ takes every value from -1 to 1. ✓ ½
$x = \pm\sqrt{2} \Rightarrow 1 - x^2 = -1; \qquad x = 0 \Rightarrow 1 - x^2 = 1$

5 So f takes every value of $\cos^{-1}$ on $[-1, 1]$. ✓ ½
$\text{Range} = [0, \pi]$

Answer **Domain** $= \left[-\sqrt{2}, \sqrt{2}\right]$**; range** $= [0, \pi]$

Watch out: Stopping at $x^2 \le 2$ and writing domain $[0, \sqrt{2}]$; negative x also work.

18 **Long answer** CBSE 2024 3 marks

Simplify: $\cos^{-1} x + \cos^{-1}\left[\frac{x}{2} + \frac{\sqrt{3 - 3x^2}}{2}\right], \frac{1}{2} \le x \le 1.$

Formula / idea: Put $x = \cos\theta$ and use $\cos(A - B) = \cos A \cos B + \sin A \sin B$.

1 Put $x = \cos\theta$. As $\frac{1}{2} \le x \le 1$: ✓ 1
$\theta = \cos^{-1} x, \qquad 0 \le \theta \le \frac{\pi}{3}$

2 Simplify the root; $\sin\theta \ge 0$ here. ✓ ½
$\sqrt{3 - 3x^2} = \sqrt{3}\sqrt{1 - \cos^2\theta} = \sqrt{3}\sin\theta$

3 Rewrite the bracket using $\cos\frac{\pi}{3} = \frac{1}{2}, \sin\frac{\pi}{3} = \frac{\sqrt{3}}{2}$. ✓ ½
$\cos\frac{\pi}{3}\cos\theta + \sin\frac{\pi}{3}\sin\theta = \cos\left(\frac{\pi}{3} - \theta\right)$

4 $\frac{\pi}{3} - \theta \in \left[0, \frac{\pi}{3}\right] \subset [0, \pi]$, so $\cos^{-1}$ and $\cos$ cancel. ✓ ½
$\cos^{-1}\left[\cos\left(\frac{\pi}{3} - \theta\right)\right] = \frac{\pi}{3} - \theta$

5 Add the two parts. ✓ ½
$\theta + \frac{\pi}{3} - \theta = \frac{\pi}{3}$

Answer $\frac{\pi}{3}$

Watch out: Writing $\cos\left(\theta - \frac{\pi}{3}\right)$ and cancelling to $\theta - \frac{\pi}{3}$; for $x > \frac{1}{2}$ that angle is negative, outside $[0, \pi]$.

19 **Long answer** CBSE 2024 3 marks

Express $\tan^{-1}\left(\dfrac{\cos x}{1-\sin x}\right)$, $-\dfrac{\pi}{2} < x < \dfrac{\pi}{2}$, in the simplest form.

Formula / idea: Write $\cos x$ and $1-\sin x$ in terms of $\dfrac{x}{2}$; then $\dfrac{1+\tan A}{1-\tan A} = \tan\left(\dfrac{\pi}{4}+A\right)$.

1 Write $\cos x$ using half angles. ✓ ½

$\cos x = \cos^2\frac{x}{2} - \sin^2\frac{x}{2} = \left(\cos\frac{x}{2} - \sin\frac{x}{2}\right)\left(\cos\frac{x}{2} + \sin\frac{x}{2}\right)$

2 Write $1-\sin x$ as a perfect square. ✓ ½

$1 - \sin x = \cos^2\frac{x}{2} + \sin^2\frac{x}{2} - 2\sin\frac{x}{2}\cos\frac{x}{2} = \left(\cos\frac{x}{2} - \sin\frac{x}{2}\right)^2$

3 Cancel the common factor $\cos\dfrac{x}{2} - \sin\dfrac{x}{2}$. ✓ ½

$$\frac{\cos x}{1-\sin x} = \frac{\cos\frac{x}{2} + \sin\frac{x}{2}}{\cos\frac{x}{2} - \sin\frac{x}{2}}$$

It is positive, since $-\dfrac{\pi}{4} < \dfrac{x}{2} < \dfrac{\pi}{4}$.

4 Divide top and bottom by $\cos\dfrac{x}{2}$. ✓ ½

$$= \frac{1+\tan\frac{x}{2}}{1-\tan\frac{x}{2}} = \tan\left(\frac{\pi}{4}+\frac{x}{2}\right)$$

5 Check the angle lies in $\left(-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right)$. ✓ ½

$$-\frac{\pi}{2} < x < \frac{\pi}{2} \;\Rightarrow\; 0 < \frac{\pi}{4} + \frac{x}{2} < \frac{\pi}{2}$$

6 So $\tan^{-1}$ and $\tan$ cancel. ✓ ½

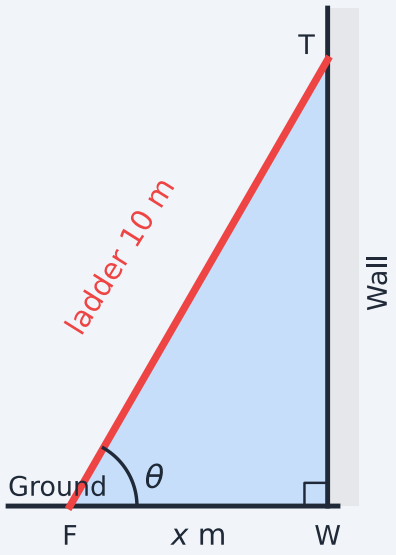
$$\tan^{-1}\left(\frac{\cos x}{1-\sin x}\right) = \frac{\pi}{4} + \frac{x}{2}$$

Answer $\dfrac{\pi}{4} + \dfrac{x}{2}$

Watch out: Skipping the range check of $\dfrac{\pi}{4} + \dfrac{x}{2}$; examiners give a mark for it.

20 **Case study** Board-pattern 4 marks

Case Study: A ladder 10 m long rests against a vertical wall. Its foot F is on level ground, x m from the foot W of the wall, and its top T touches the wall. The angle θ that the ladder makes with the ground is given by $\theta = \cos^{-1}\left(\frac{x}{10}\right)$.



(i) Find θ when $x = 5$ m. 1 mark

(ii) Write the domain of the function $f(x) = \cos^{-1}\left(\frac{x}{10}\right)$ and the range of its principal value branch. 1 mark

(iii) The foot of the ladder slips away from the wall until $\theta = \frac{\pi}{6}$. Find the new distance x of the foot from the wall and the height WT at which the top now touches the wall. 2 marks

Formula / idea: In right triangle FWT: $\cos\theta = \frac{\text{FW}}{\text{FT}} = \frac{x}{10}$ and $\sin\theta = \frac{\text{WT}}{\text{FT}} = \frac{\text{WT}}{10}$.

(i) 1 mark

1 Put $x = 5$.
$\theta = \cos^{-1}\left(\frac{5}{10}\right) = \cos^{-1}\left(\frac{1}{2}\right)$ ✓ ½

2 $\cos\frac{\pi}{3} = \frac{1}{2}$ and $\frac{\pi}{3} \in [0, \pi]$.
$\theta = \frac{\pi}{3}$ ✓ ½

Answer (i) $\theta = \frac{\pi}{3}$

(ii) 1 mark

1 The input of $\cos^{-1}$ must lie in $[-1, 1]$.
$-1 \le \frac{x}{10} \le 1$

2	Multiply each part by 10. $-10 \le x \le 10 \;\Rightarrow\; \text{Domain} = [-10, 10]$	✓ ½
3	Values of $\cos^{-1}$ lie in its principal value branch. $\text{Range} = [0, \pi]$	✓ ½

Answer (ii) **Domain $[-10, 10]$; range $[0, \pi]$**

(iii)

2 marks

1	Put $\theta = \dfrac{\pi}{6}$ in $\cos\theta = \dfrac{x}{10}$. $\dfrac{x}{10} = \cos\dfrac{\pi}{6} = \dfrac{\sqrt{3}}{2}$	✓ ½
2	Solve for x. $x = 5\sqrt{3}\ \text{m} \approx 8.66\ \text{m}$	✓ ½
3	For the height use $\sin\theta = \dfrac{\text{WT}}{10}$. $\text{WT} = 10\sin\dfrac{\pi}{6} = 10 \times \dfrac{1}{2}$	✓ ½
4	Hence the top touches the wall at: $\text{WT} = 5\ \text{m}$	✓ ½

Answer (iii) **New distance $x = 5\sqrt{3} \approx 8.66$ m; height WT $= 5$ m**

Watch out: In (ii), writing the domain as $[-1, 1]$; that is the condition on $\dfrac{x}{10}$, not on x.